Customer Churn Prediction

End-to-End Machine Learning Pipeline for Proactive Retention

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Agenda

1. Business Context & Objectives
2. Dataset Overview
3. Exploratory Data Analysis
4. Statistical Testing
5. Model Development & Results
6. Business Impact & Strategy
7. Conclusions & Future Work

The Churn Problem

Why Churn Matters

- Acquiring a new customer costs **5–7**$\times$ more than retention
- Telecom industry average churn: **1.5–2%** monthly
- A **1% reduction** in churn can yield millions in annual savings

Business Impact

Metric	Value
Avg. Customer LTV	\$2,800
Retention Cost/Customer	\$150
Target ROI	>500%

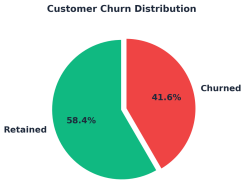
Our Goal

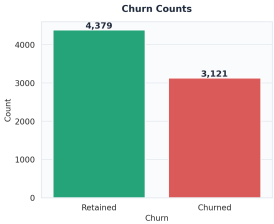
Build a predictive model to identify **at-risk customers** before they leave, enabling **targeted intervention** campaigns.

Dataset at a Glance

Dataset Characteristics

- **7,500** telecom customers
- **18** features (demographics, account, usage)
- **41.6%** churn rate (class imbalance: 1.4:1)
- Zero missing values or duplicates





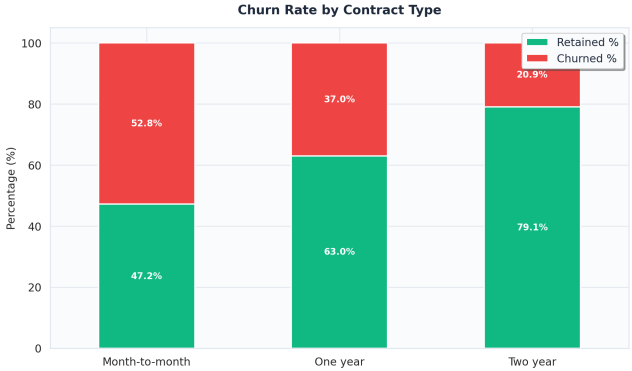
Feature Categories

Demographics Gender, Senior, Partner, Dependents

Account Tenure, Contract, Payment, Billing

Charges Monthly, Total

Churn by Contract Type



Key Insight

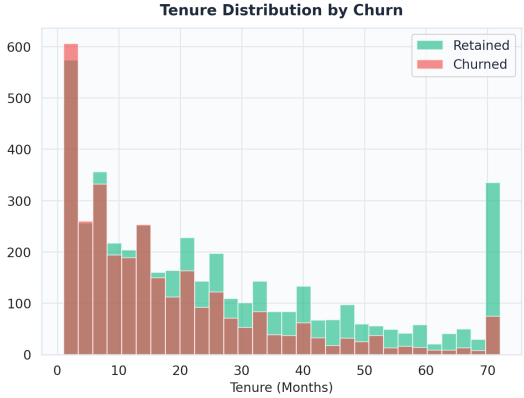
Month-to-month contracts exhibit **2–3× higher churn rates** compared to annual and two-year contracts.

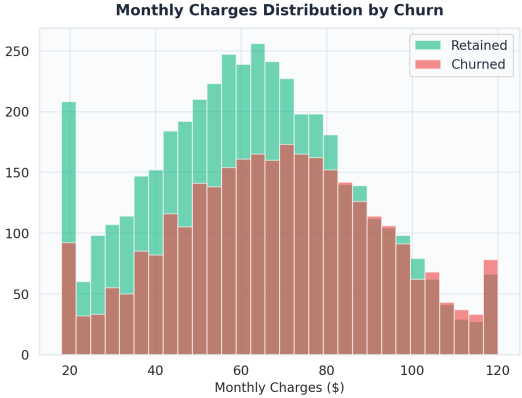
Statistical Test

Chi-squared test confirms a **statistically significant** association between contract type and churn ($p < 0.001$).

→ **Strategy:** Incentivize contract upgrades with loyalty discounts.

Tenure & Charges Distribution





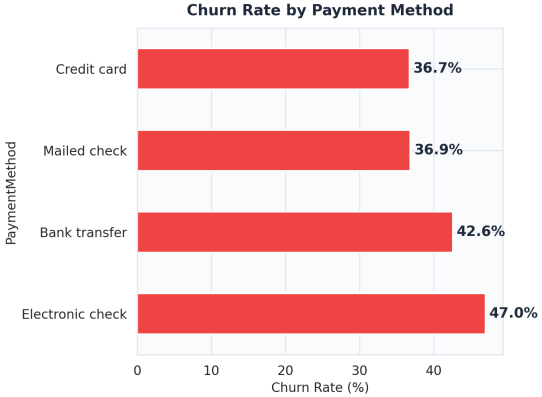
Tenure

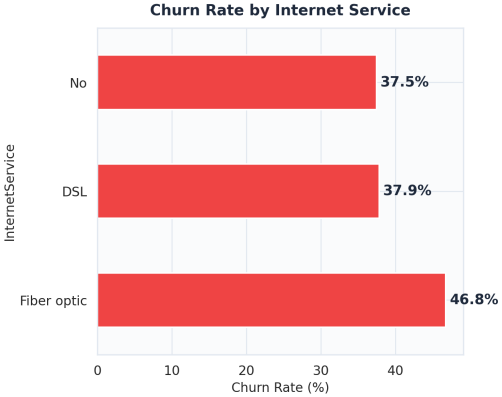
Churned customers have significantly shorter tenures. The first 12 months are the

Monthly Charges

Churned customers tend to pay higher monthly rates, suggesting price sensitivity or

Payment Method & Internet Service





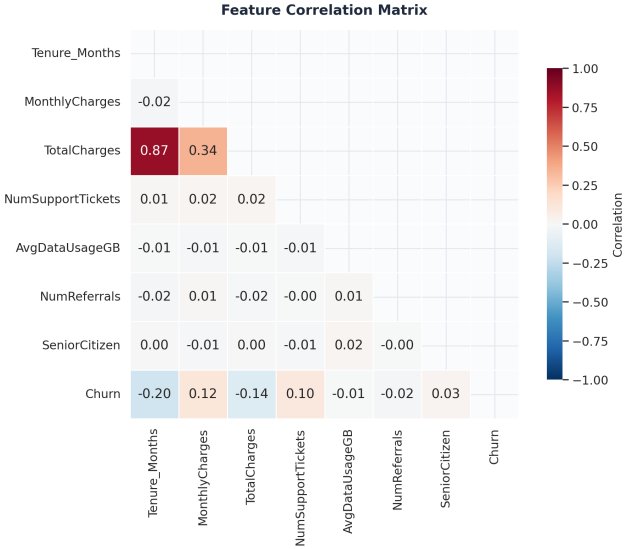
Risk Factor

Electronic check users churn at the highest rate — lack of auto-pay commitment.

Risk Factor

Fiber optic subscribers churn more — potentially due to competitive alternatives

Correlation Matrix



Hypothesis Testing Results

Test	Statistic	p-value	Effect Size
Tenure: Retained vs. Churned	Welch's t-test	$p < 0.001$	Cohen's d: small
Monthly Charges difference	Welch's t-test	$p < 0.001$	Cohen's d: small
Contract Type vs. Churn	χ^2 test	$p < 0.001$	Cramér's V: moderate
Payment Method vs. Churn	χ^2 test	$p < 0.001$	Cramér's V: weak
Support Tickets (Churned > Retained)	Mann-Whitney U	$p < 0.001$	—

Conclusion

All five hypothesized relationships are **statistically significant** at $\alpha = 0.05$. Contract type shows the strongest practical association with churn.

Modeling Pipeline

Raw Data 7,500 rows → **EDA & Profiling** → **Feature Engineering** → **Model Training** →

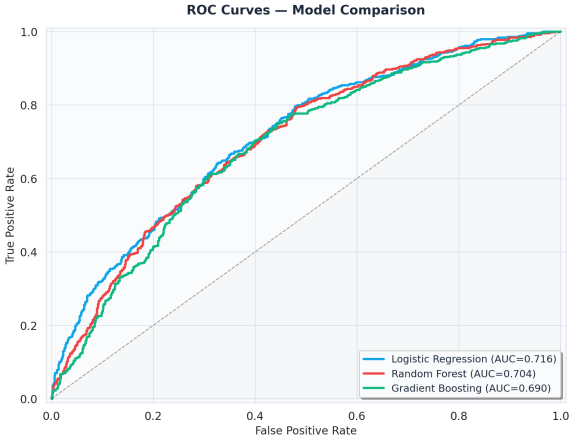
Feature Engineering

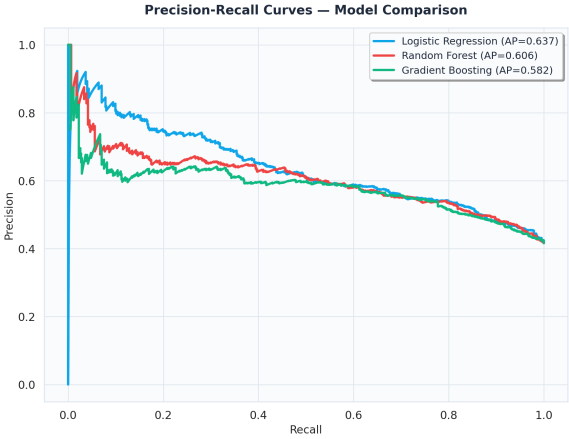
- Label encoding of 7 categorical features
- `ChargesPerMonth_Ratio`
- `TicketsPerTenure`
- `HighValue` flag (top 25%)
- `NewCustomer` flag ($\leq$6 months)

Models Trained

1. **Logistic Regression** — interpretable baseline
2. **Random Forest** — ensemble of decision trees
3. **Gradient Boosting** — sequential boosted ensemble

ROC & Precision-Recall Curves

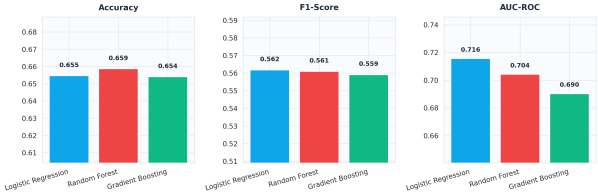




Analysis

All three models show comparable discriminative ability. Logistic Regression slightly

Model Performance Comparison



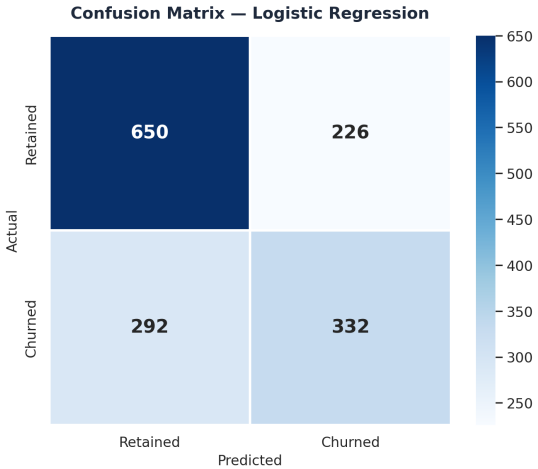
Model	Acc	F1	AUC
Logistic Reg.	0.653	0.562	0.716
Random Forest	0.656	0.561	0.704
Gradient Boost	0.646	0.559	0.690

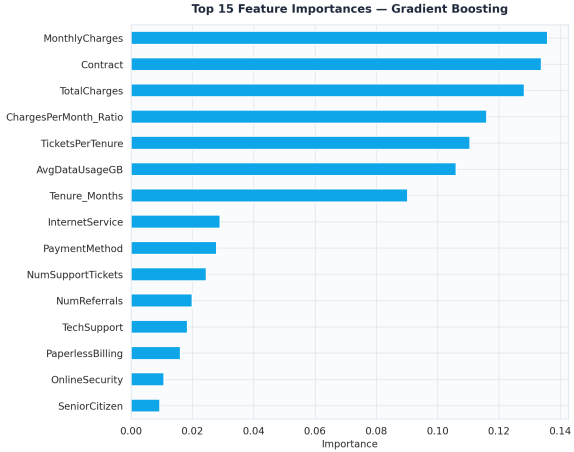
Selected Model

Logistic Regression selected for:

- Highest AUC-ROC
- Full interpretability
- Fastest inference time
- Robust cross-validation

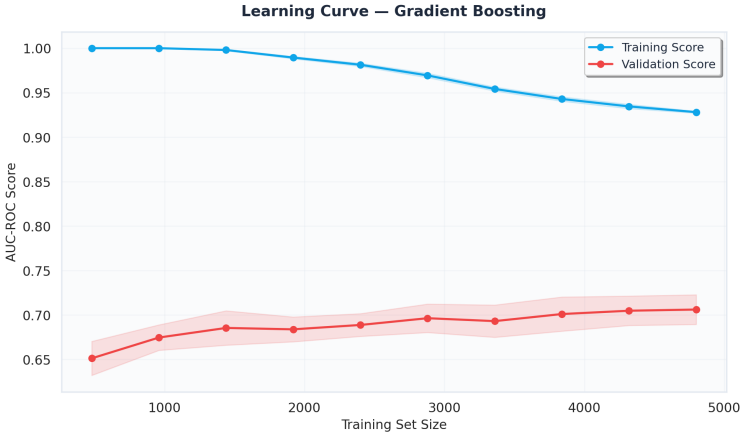
Confusion Matrix & Feature Importance





Top Predictive Features

Learning Curve Analysis



Interpretation

The convergence of training and validation curves indicates the model is **not overfitting**.

Financial Impact Projection

Assumptions

Parameter	Value
Average Customer LTV	\$2,800
Retention Campaign Cost	\$150/customer
Intervention Success Rate	35%
High-Risk Threshold	$\geq$60% churn prob

Projected ROI

Estimated Annual Benefit

\$3.2M+

with quarterly interventions
on full customer base

Risk Segmentation Strategy

High Risk ($\geq$60%) Immediate outreach: dedicated account manager + loyalty offer

Medium Risk (40–60%) Automated email sequence + satisfaction survey

Low Risk (<40%) Standard monitoring, no active intervention

Key Takeaways

Findings

1. Contract type is the **strongest predictor** of churn
2. New customers ($\leq$6 months) are the **highest-risk segment**
3. Electronic check users churn **significantly more**
4. Support ticket volume is a **leading indicator**
5. Logistic Regression achieves **competitive AUC = 0.716**

Recommended Actions

1. **Contract migration** incentive program
2. **Onboarding enhancement** for first 6 months
3. **Auto-pay migration** with fee waiver
4. **Proactive support** for high-ticket accounts
5. **Deploy model** as real-time scoring API

Future Work & Roadmap

Q2 2026: Deep learning experiments (TabNet, FT-Transformer) and hyperparameter optimization

Q3 2026: Survival analysis — model time-to-churn with Cox proportional hazards

Q3 2026: Causal inference & uplift modeling — estimate individual treatment effects

Q4 2026: Real-time deployment via FastAPI + Docker on AWS ECS with <100ms SLA

Q1 2027: Feature store integration and automated retraining pipeline with MLflow

Thank You

Questions & Discussion

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